COMPUTING WITH TILES

Wang tiles that add, and a wall that counts in Fibonacci numbers

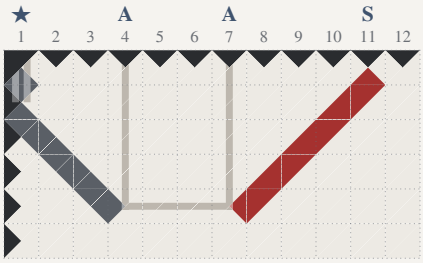
I. WANG TILES

A *Wang tile* is a square with a type on each edge, shown here by color. Tiles are laid edge to edge so that touching edges match, and may not be rotated or reflected. Hao Wang (王浩)[†] conjectured in 1961 that any set of such tiles that can cover the plane can also cover it periodically. Robert Berger (1966) showed otherwise: tiles can mimic any computer program (they are *Turing complete*), so whether a set tiles the plane is undecidable. Along the way he found the first *aperiodic* set, 20,426 tiles that tile the plane but never repeat. At right is the smallest aperiodic set possible: 11 tiles with 4 edge types (Jeandel & Rao, 2015).



a minimal aperiodic set of tiles

II. TILES THAT ADD

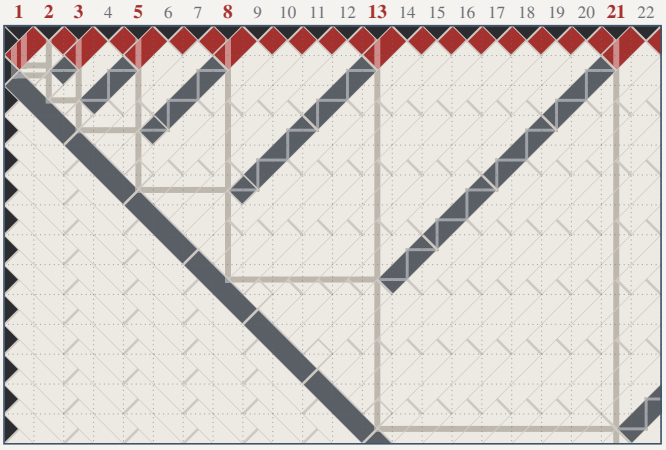


A at 4 and 7 force S to 11

Tiles can do arithmetic because the forced tiling carries information. This tiling fills a quarter plane whose top and left edges must be black. Only ★ fits in the corner, place the **A** tiles at your input values, and the rest follows. In these 16 tiles (after Grünbaum & Shephard) the edge types draw lines. A thick ray runs down from each **A**; a charcoal diagonal leaves ★ at 45°; where it meets the first ray it turns and runs horizontally until it meets the second ray, and a red diagonal climbs back from there. The diagonals must be equal, so the climb reaches the top at column $A_1 + A_2$ where **S** must be placed.



III. FIBONACCI NUMBERS



as it could be laid in 2″ × 8″ subway tile

Run the adding machine over and over and the tiles generate the Fibonacci numbers, each the sum of the two before.[‡] The **F′** and **F″** tiles in the corner mark the first two numbers, 1 and 2. Every **F** sends a thick ray straight down. A long charcoal diagonal leaves the corner at 45°, and where it meets the ray of the **F** at column n, at depth n, the ray turns and runs right to the ray of the next **F**, at column m. There a charcoal staircase climbs back at 45°, rising n rows as it moves n columns right, and reaches the top at column $m + n$: the next **F**, which then sends its own ray down.

Our goal was to achieve an aperiodic tiling for the home using tiles in easily obtainable practical shapes. Wang tiles are attractive because the aperiodicity is from edge matching rather than shape. This tiling achieves our goal with a more pleasant appearance than the Jeandel & Rao tile set.



These 18 tiles have thirteen edge types; as usual edges must match. With the top and left edges black they fill the quarter plane in exactly one way, and since the Fibonacci numbers spread ever further apart, that tiling is aperiodic. **F′** and **F″** fix the start: add two tiles that let **F″** stand at column 3 instead, and the same machinery produces the Lucas numbers 1, 3, 4, 7, 11, 18, 29.

[†] In Cantonese, Wang's surname 王 is a homophone of 黃, Wong: a name held by several members of this home.

[‡] Grünbaum & Shephard, Fig. 11.4.2 has three misprints: its list of tiles leaves out the one that continues the diagonal, its example drops two of the "6" edge labels in column 3, and its caption counts 11 among the Fibonacci numbers.

Addition and Fibonacci tiles after Grünbaum & Shephard, Tilings and Patterns, §11.4.

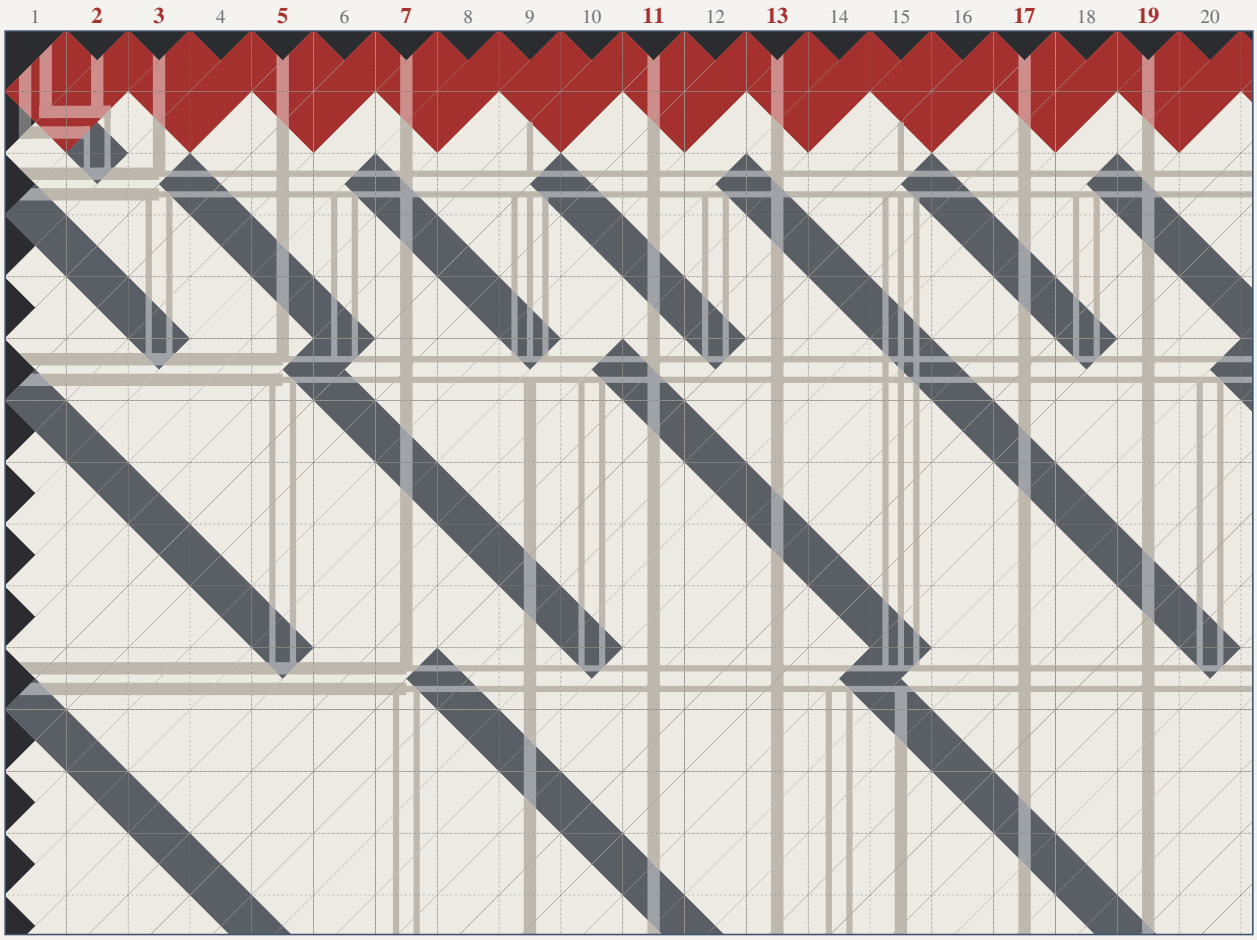
C. Scott Ananian · October 2026

A WALL THAT FINDS PRIMES

Wang tiles that sieve for primes, in ordinary 2″ × 8″ subway tile

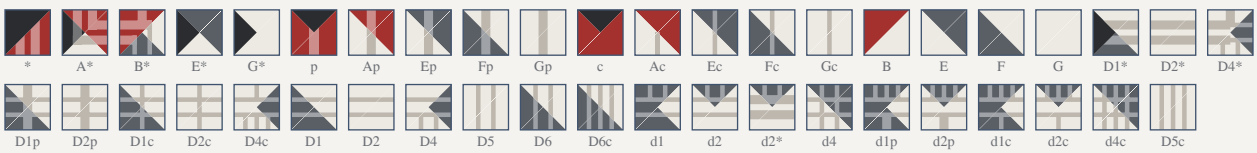
IV. A PRIME NUMBER SIEVE

The wall in this room is an aperiodic tiling with a set of 43 Wang tiles. Each edge type is shown by a tile color plus a pattern of etched lines, and a run of matching edges is laid as a single subway tile set on the diagonal. Column *n* stands for the number *n*. The double lines and charcoal diagonals create rows of square boxes with odd sides, repeating across the wall and growing going down: 3, 5, 7, 9, …. The first box in each row, drawn with thick double lines, sends a thick single ray up toward the top. If the ray reaches the top without hitting the vertical edge of a box above it, it marks a prime; if it hits an edge, it turns into a thin line marking a composite. (Even numbers get no ray; 2 has a tile of its own, **B***.) The thick rays reach the red top row at 2, 3, 5, 7, 11, 13, 17 and 19.



the Wang tiling of the wall: columns 1–20, top 15 rows

V. THE 43 TILES



The 43 tiles have 15 edge types; edges must match, and top and left edges must be black. Given these constraints, the tiles fill the infinite quarter plane in exactly one way, and that tiling is aperiodic since its boxes keep growing and the primes never repeat.

Tile set PRIME2b from github.com/cscott/tile-hack (wangprime.js).

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